

Motion in a Straight Line ✍



- ① A vehicle travelled half the distance with speed v and the remaining distance with speed $2v$. Its average speed is -
- a. $\frac{2v}{3}$ ~~b. $\frac{4v}{3}$~~ c. $\frac{3v}{4}$ d. $\frac{v}{3}$

$\frac{S}{v}$ $t_1 = \frac{S}{v}$	$\frac{S}{2v}$ $t_2 = \frac{S}{2v}$
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$$\begin{aligned} \text{avg speed} &= \frac{S+tS}{t_1+t_2} \\ &= \frac{2S}{\frac{S}{v} + \frac{S}{2v}} = \frac{2S}{\frac{S}{v}(1+\frac{1}{2})} \\ &= \frac{2v}{\frac{3}{2}} = \frac{4v}{3} \text{ (Ans)} \end{aligned}$$

- ② If the velocity of a particle is $v = At + Bt^2$, where A and B are constants, then the distance travelled by it between 1 sec to 2 sec is -
- a. $3A + 7B$ ~~b. $\frac{3}{2}A + \frac{7}{3}B$~~ c. $\frac{A}{2} + \frac{B}{3}$ d. $\frac{3}{2}A + 4B^2$

$$\begin{aligned} S &= \int_1^2 v dt = \int_1^2 (At + Bt^2) dt = A \int_1^2 t dt + B \int_1^2 t^2 dt \\ &= \left[A \frac{t^2}{2} + B \frac{t^3}{3} \right]_1^2 = A \frac{2^2}{2} + B \frac{2^3}{3} - A \frac{1}{2} - B \frac{1}{3} \\ &= 2A + \frac{8B}{3} - \frac{A}{2} - \frac{B}{3} \\ &= \frac{3A}{2} + \frac{7B}{3} \text{ (Ans)} \end{aligned}$$

$\int x^n dx = \frac{x^{n+1}}{n+1}$

- ③ A car runs at a constant speed on a circular track of radius 100 m, taking 62.8 sec for every circular lap. The average velocity and the average speed for each circular lap respectively is -
- a. 10 m/s, 0 b. 0, 0 ~~c. 0, 10 m/s~~ d. 10 m/s, 100 m/s



here avg velocity is 0 m/s but avg speed is 10 m/s, so, the correct option is 0, 10 m/s

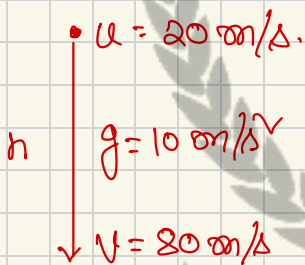
- ④ A particle moves along a straight line such that its displacement at any time t is given by $s = 3t^3 + 7t^2 + 14t + 5$. The acceleration of the particle at $t = 1 \text{ sec}$ is -
- a. 18 m/s^2 ~~b. 32 m/s^2~~ c. 29 m/s^2 d. 24 m/s^2

$$s = 3t^3 + 7t^2 + 14t + 5$$

$$\frac{ds}{dt} = 9t^2 + 14t$$

$$\frac{d^2s}{dt^2} = 18t + 14 = 18 \times 1 + 14 = 32 \text{ m/s}^2 \text{ (Ans)}$$

- ⑤ A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s . The height of the tower is 300 m .



$$v^2 = u^2 + 2gh$$

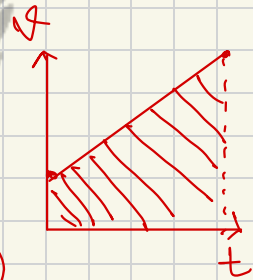
$$80^2 = 20^2 + 2 \times 10 \times h$$

$$(80^2 - 20^2) = 20h$$

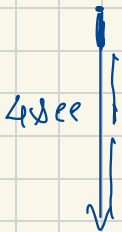
$$20h = (80+20)(80-20)$$

$$20h = 100 \times 60$$

$$\therefore h = 300 \text{ m (Ans)}$$



⑥



1st, 2nd, 3rd and 4th sec.

$$\text{displacement} = 1:3:5:7$$

- ⑦ A ball is dropped from a high rise platform at $t=0$ starting from rest. After 6 sec, another ball is thrown downward from the same platform with a speed u . The two balls meet at $t=18$ sec. What is the value of u ?

a. 75 m/s b. 55 m/s c. 40 m/s d. 60 m/s

$u=0$ $t=0$
 h
 $t=18$ sec.
 $h = \frac{1}{2} g t^2$
 $= \frac{1}{2} g \times 18^2$

u
 h
 12 sec
 $h = u \times 12 + \frac{1}{2} g 12^2$

$$\therefore \frac{1}{2} g 18^2 = 12u + \frac{1}{2} g 12^2 \quad \left| \quad v = \frac{S}{T} (18+12)(18-12) \right.$$

$$5 \times 18^2 - 5 \times 12^2 = 12u$$

$$5(18^2 - 12^2) = 12u$$

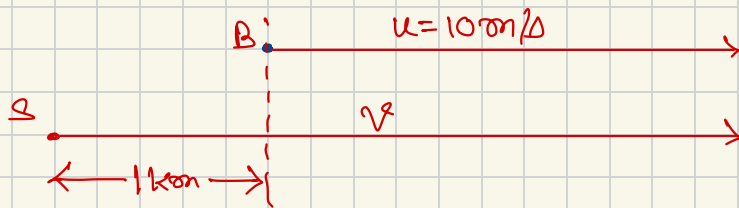
$$= \frac{5}{12} \times 20^2 \times 6$$

$$= 75 \text{ m/s (Ans)}$$

- ⑧ What will be the ratio of the distance covered by a freely falling body from rest in 4th and 5th second of journey. 7:9

1:3:5:7:9:11

9. A bus is moving with a speed of 10 m/s on a straight road. A scooterist wishes to overtake the bus in 100 sec. If the bus is at a distance of 1 km from the scooterist, with what speed should the scooterist chase the bus?



$$t = 100 \text{ sec.}$$

$$\frac{(u \times 100)}{1000} + 1 = \frac{(v \times 100)}{1000}$$

$$\frac{1000}{1000} + 1 = \frac{v}{10}$$

$$2 = \frac{v}{10} \quad \therefore v = 20 \text{ m/s. (Ans)}$$